STUDENT PERFORMANCE DATA SCIENCE

1. Data Collection

Attendance Records
Assignment Scores
Results
Internal Marks
Final Exam

2. Data Integration

- Combine records using Student ID
- Align formats and time periods

3. Data Understanding

- Inspect fields
- Identify data types
- Check ranges

4. Data Cleaning

- Handle missing marks/attendance
- Remove duplicate student records
- Correct invalid values

5. Data Transformation

- Standardize column names/formats
- Encode categories; scale scores
- Create total/average features

6. Analysis-Ready Dataset

- Structured student-level table for performance analysis

Student Performance Analysis

HOSPITAL DATA FOR ANALYSIS

PATIENT DATA SOURCES

Registration Forms

- Patient ID
- Age and contact details
- Demographics

Laboratory Reports

- Blood tests and values
- Reference ranges
- Test date and time

Medical Devices

- Vital signs
- Heart rate and blood pressure
- Oxygen level (SpO_2)

Doctor Notes

- Symptoms reported
- Diagnosis and treatment plan
- Clinical observations

DATA PREPARATION

- Standardize Formats
- Match Patient Identifiers
- Check Missing / Duplicate Records
- Protect Patient Privacy

INTEGRATED PATIENT DATASET

ANALYSIS USES

- Identify Patterns
  Common diseases and trends
- Support Clinical Decisions
  Treatment selection support
- Monitor Patient Status
  Risk alerts and vital changes

DATA CLEANING FOR ONLINE SHOPPING

Raw E-commerce Dataset
Customer details • order records • product entries • purchase amounts

Missing Customer Details

Impute or mark unknown; verify required fields

Duplicate Orders

Detect repeated order IDs; remove duplicates

Wrong Product Entries

Validate product IDs/categories; correct inconsistencies

Unusual Purchase Amounts

Check outliers; verify genuine transactions vs data errors

VALIDATION CHECKS
Completeness • uniqueness • consistency • valid ranges

Clean Dataset Ready for Analysis

SMART CITY TRAFFIC DATA INTEGRATION

DATA SOURCES

CCTV Cameras
Vehicle counts
Traffic density

GPS Devices
Location and speed trajectories

Road Sensors
Flow, occupancy signal status

Mobile Apps
Routes, reports travel-time updates

INTEGRATION PIPELINE

1. Standardize
Common field names, units and timestamp formats

2. Align and Match
Synchronize time; match common road/location IDs

3. Merge and Validate
Join records; remove duplicates and check missing values

UNIFIED TRAFFIC DATASET

Integrated Records
One consistent table by road and time

Analysis and Decisions
Congestion detection
Signal timing • Route planning

RESULT: one integrated dataset supports city-wide traffic decisions

BANK LOAN DATA TRANSFORMATION

INPUT CUSTOMER DATA

- Income
- Age
- Job Type
- Loan Amount
- Credit Score

TRANSFORMATION PIPELINE

1. Encode Categories

Convert job type and other labels into numeric features

2. Normalize Values

Scale selected values to a common range (e.g., 0–1)

3. Standardize / Scale

Use mean and standard deviation when appropriate

4. Check Data Quality

Handle missing values, outliers and inconsistent units

TRANSFORMED FEATURE MATRIX

Model Training

Loan approval prediction

Evaluation

Measure performance on held-out data

Avoid data leakage: fit transformations on training data only.